

# MATH 345 Skeleton Notes

## Unit 4: Orthogonality and least squares

### Section 4.4: \$QR\$ factorization and least squares

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#### QR factorization

Recall that a list of vectors is orthonormal when the vectors have length 1 and are pairwise orthogonal. Also recall the notion of an upper triangular matrix from Triangular and Diagonal Matrices.

**Definition** Let  $A$  be an  $m \times n$  matrix with independent columns. A

**QR factorization** of  $A$  is an  $m \times n$  matrix  $Q$ , with orthonormal columns, and an invertible upper triangular  $n \times n$  matrix  $R$ , with positive entries on its diagonal, so that  $A = QR$ .

**Warning: Rectangular  $Q$**  In a  $QR$  factorization of an  $m \times n$  matrix with  $m > n$ , the matrix  $Q$  has orthonormal columns, so  $Q^T Q = I_n$ . But  $Q$  is not square, and  $QQ^T$  is not usually  $I_m$ .

Normal equations are conceptually important, but  $QR$  is often a better computational method. If  $A = QR$  with  $Q$  having orthonormal columns, then

$$\|Ax - b\| = \|QRx - b\|.$$

Orthogonality reduces the least-squares problem to

$$R\hat{x} = Q^T b,$$

which is triangular and easier to solve accurately.

### Activity: Reading QR least-squares code

The following code uses  $QR$  to solve the same least-squares problem from the Unit 2 target activity.

```
A = np.array([
    [1.0, 0.0],
    [0.0, 1.0],
    [1.0, 1.0],
])
b = np.array([1.0, 2.0, 4.0])

Q, R = np.linalg.qr(A)
x_qr = np.linalg.solve(R, Q.T @ b)
x_lstsq = np.linalg.lstsq(A, b, rcond=None)[0]

Q.T @ Q, x_qr, x_lstsq
```

1. What should  $Q.T @ Q$  be close to?
2. Why does the code solve a system with  $R$  instead of a system with  $A$ ?
3. Why should  $x\_qr$  and  $x\_lstsq$  agree?
4. Does this mean  $Ax = b$  has an exact solution?

### Activity

Let

$$Q = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \end{bmatrix}.$$

Check that the columns are orthonormal and compute  $Q^T Q$ .

### Activity

Solve the linear system  $Rx = \mathbf{b}$  where

$$R = \begin{bmatrix} 4 & 2 & 7 \\ 0 & 3 & 5 \\ 0 & 0 & 2 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 30 \\ 21 \\ 6 \end{bmatrix}$$

The QR factorization is a matrix version of the *Gram-Schmidt orthogonalization process*, which has many applications in many numerical algorithms.

**Theorem** Every  $m \times n$  matrix  $A$  with linearly independent columns has a QR factorization  $A = QR$ .

**Proof** Let  $A = [\mathbf{c}_1 \mathbf{c}_2 \cdots \mathbf{c}_n]$  be the matrix with columns  $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n$ . Apply Gram-Schmidt process to obtain orthogonal vectors  $\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_n$  where:  $\mathbf{f}_1 = \mathbf{c}_1$  and

$$\mathbf{f}_k = \mathbf{c}_k - \frac{\mathbf{c}_k \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 - \frac{\mathbf{c}_k \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} \mathbf{f}_2 - \cdots - \frac{\mathbf{c}_k \cdot \mathbf{f}_{k-1}}{\|\mathbf{f}_{k-1}\|^2} \mathbf{f}_{k-1}$$

for  $k = 2, 3, \dots, n$ .

Let  $\mathbf{q}_k = \frac{\mathbf{f}_k}{\|\mathbf{f}_k\|}$  for each  $k$ . Then  $\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_n$  are orthonormal vectors.

We can rewrite each column  $\mathbf{c}_k$  as a linear combination of the  $\mathbf{q}_i$ :  $\mathbf{c}_1 = \|\mathbf{f}_1\| \mathbf{q}_1$ ,

$$\mathbf{c}_2 = (\mathbf{c}_2 \cdot \mathbf{q}_1) \mathbf{q}_1 + \|\mathbf{f}_2\| \mathbf{q}_2$$

$$\mathbf{c}_3 = (\mathbf{c}_3 \cdot \mathbf{q}_1) \mathbf{q}_1 + (\mathbf{c}_3 \cdot \mathbf{q}_2) \mathbf{q}_2 + \|\mathbf{f}_3\| \mathbf{q}_3$$

and so on. This gives us the matrix factorization:

$$\begin{aligned} A &= [\mathbf{c}_1 \mathbf{c}_2 \cdots \mathbf{c}_n] \\ &= [\mathbf{q}_1 \mathbf{q}_2 \cdots \mathbf{q}_n] \begin{bmatrix} \|\mathbf{f}_1\| & \mathbf{c}_2 \cdot \mathbf{q}_1 & \mathbf{c}_3 \cdot \mathbf{q}_1 & \cdots & \mathbf{c}_n \cdot \mathbf{q}_1 \\ 0 & \|\mathbf{f}_2\| & \mathbf{c}_3 \cdot \mathbf{q}_2 & \cdots & \mathbf{c}_n \cdot \mathbf{q}_2 \\ 0 & 0 & \|\mathbf{f}_3\| & \cdots & \mathbf{c}_n \cdot \mathbf{q}_3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \|\mathbf{f}_n\| \end{bmatrix} \end{aligned}$$

So  $A = QR$  where  $Q = [\mathbf{q}_1 \mathbf{q}_2 \cdots \mathbf{q}_n]$  has orthonormal columns and  $R$  is upper triangular with positive diagonal entries.

### Activity

Consider the matrix

$$A = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix}.$$

Find its QR factorization.

### Activity

Find the QR factorization of the matrix

$$A = \begin{bmatrix} 6 & 2 \\ 0 & 3 \end{bmatrix}.$$

**Theorem** Every square, invertible matrix  $A$  can be written as  $A = QR$ , where  $Q$  is an orthogonal matrix, and  $R$  is an upper triangular matrix with positive diagonal entries.